



What should be added/subtracted to
each term of **$a:b$** to convert them to
 $c:d$?

Que: What should be added to each term of $9 : 16$ so that new ratio becomes $2 : 3$?

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$$\frac{9+5}{16+5} \rightarrow \frac{2}{3}$$

$$\frac{9+y}{16+y} \neq \frac{2}{3}$$

$$3(9+y) = 2(16+y)$$

$$27+3y = 32+2y$$

$$\boxed{y = 5}$$

Que: What should be subtracted from each term of $15 : 19$
so that new ratio becomes $3 : 4$?

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$$\frac{3}{4} \leftarrow \frac{15}{19} \rightarrow \frac{3}{4}$$

$$\frac{15-y}{19-y} \rightarrow \frac{3}{4}$$

$$4(15-y) = 3(19-y)$$

$$60 - 4y = 57 - 3y$$

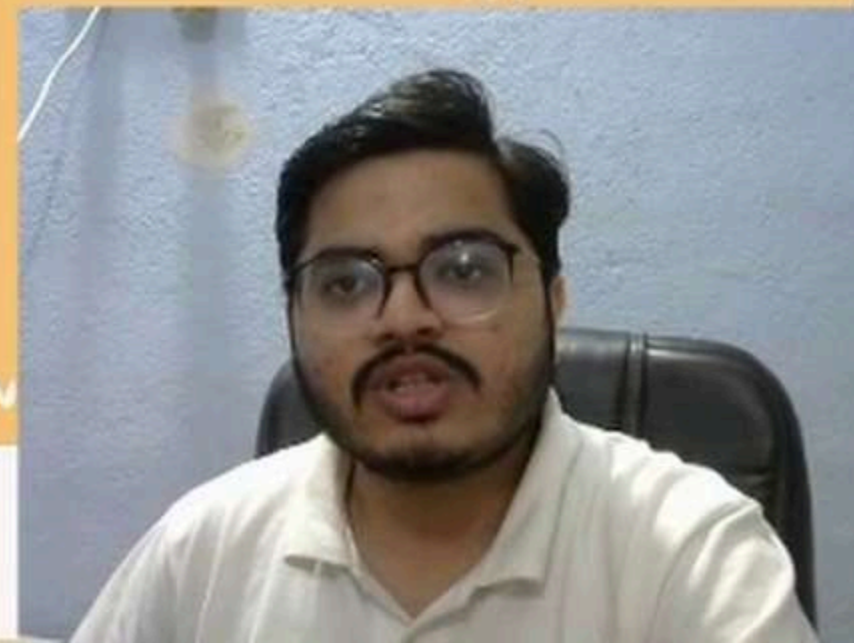
$$\boxed{3 = y}$$

Que: What should be added to each term of $11 : 21$ so that new ratio becomes $3 : 5$?

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Que: What should be added to each term of 11 : 21 so that new ratio becomes 3 : 5?

$$\frac{8}{2} = 4$$

$$\frac{11}{21} \rightarrow \frac{3}{5}$$

$$11 * 5 = 55$$
$$21 * 3 = 63$$

$$\frac{18}{25}$$

$$\frac{3}{5}$$

$$5 - 3 = 2$$

Que: What should be subtracted from each term of $23 : 38$
so that new ratio becomes $4 : 7$?



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Que: What should be subtracted from each term of $23 : 38$ so that new ratio becomes $4 : 7$?

$$23 \times 7 \rightarrow 161$$
$$38 \times 4 \rightarrow 152$$

$$\frac{23}{38}$$

$$\frac{4}{7}$$

$$7 - 4$$

$$\frac{9}{3} = 3$$

$$\frac{23-9}{38-12} = \frac{14}{26}$$